
Discussion of

Do Firm Characteristics Predict Industrial Production Growth?

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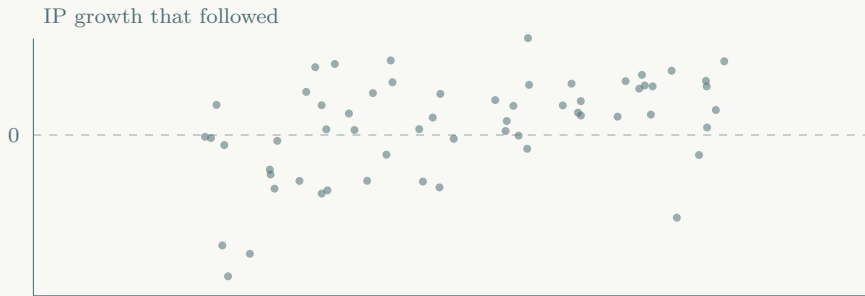
What the paper estimates

$$Q_{\tau}(y_{t+h} \mid X_t) = f(X_t)$$

Q_{τ}	conditional τ -quantile; $\tau = 0.05$ is “growth-at-risk”
y_{t+h}	annualized industrial production (IP) growth from month t to $t + h$
X_t	108 firm characteristics (Jensen, Kelly and Pedersen 2023), averaged across firms, plus lagged IP growth
f	conditional quantile function, estimated by a quantile random forest
SHAP	Shapley additive explanations: splits each forecast across the inputs

The paper estimates a forecast, then concludes that the characteristics **proxy for financial conditions, inform policy, and reveal the drivers of growth-at-risk.**

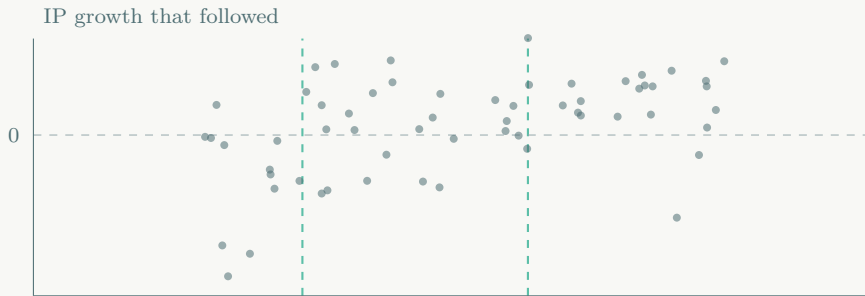
How the forest forecasts: one tree, one input



Each dot is a past month: its input, and the IP growth that followed.

Illustration with made-up data. The paper uses 500 trees, each on a random subsample, and 109 inputs.

How the forest forecasts: one tree, one input

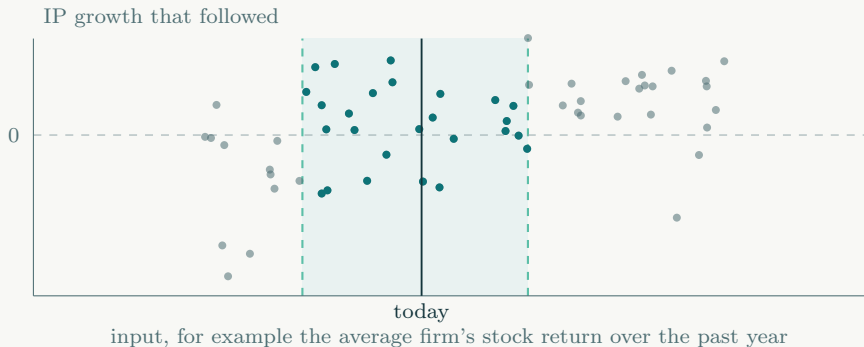


input, for example the average firm's stock return over the past year

The tree splits the months by the input, so growth within each group is as alike as possible.

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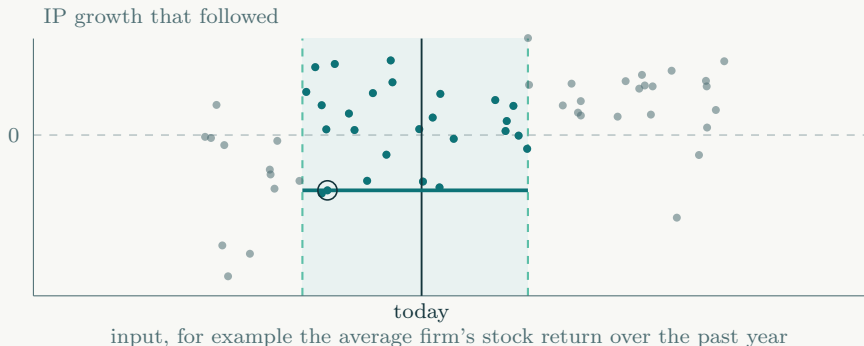
How the forest forecasts: one tree, one input



Today's input falls in one group. Those past months count as similar.

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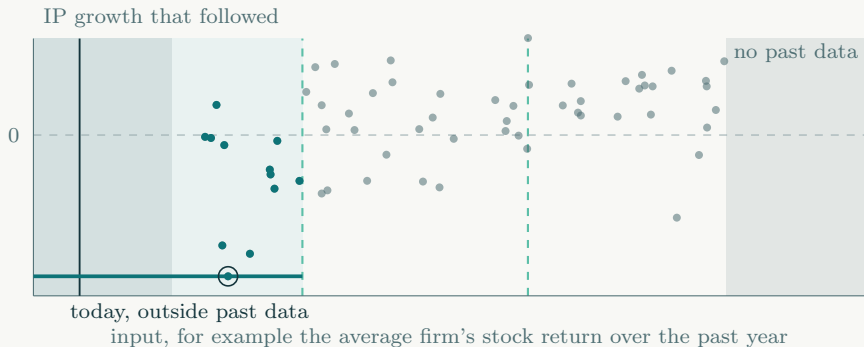
How the forest forecasts: one tree, one input



The forecast is the 5th percentile of what followed them: here, the second-worst of 26 months.

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How the forest forecasts: one tree, one input



An input beyond past experience gets the edge group's forecast: **the forest cannot extrapolate.**

Illustration with made-up data. The paper uses 500 trees, each on a random subsample, and 109 inputs.

f does not identify mechanisms or shocks.
A structural model can.

	f and SHAP	A structural model
Mechanism	the same f is consistent with financing constraints, demand news, or risk premia	each mechanism predicts which firms respond and which variables move first
Shocks	splits the forecast across the inputs	splits IP growth across credit, demand and risk shocks

The paper calls its top predictors “plausible proxies for financial conditions.”
Neither the forecast nor SHAP can test that.

Evaluating a new policy rule needs a structural model

$$f = f_\psi \quad \psi: \text{policy rule}$$

Lucas (1976)

- The sample spans pre-Volcker, Volcker–Greenspan and zero-bound policy.
- Under a new rule the true f can change; past data cannot show how.
- A structural model fixes preferences, technology and frictions, and solves for the new f .
- The paper says its forecast helps policy even at long horizons.
At 24 months, the historical 5th percentile beats its growth-at-risk forecast.

A predictor of growth-at-risk need not drive it

- The paper sets out to find “the drivers of growth-at-risk” (Section 6).
- Top inputs include stock returns, which fall when investors expect a downturn.
- Predicting low growth says nothing about the effect of higher stock prices.
- Calling an input a driver needs exogenous variation or a structural model.
The paper has neither.
- An identified structural model can separate causes from signals.

As a forecasting exercise, the evidence is weak

- No factor-model benchmark, and no model that combines X with FRED-MD.
- Several top predictors are past stock returns (Fama 1990).
What do the accounting variables add beyond them?
- All results include April 2020, when IP fell more than 10% in one month.
Show them without 2020.
- Giglio, Kelly and Pruitt (2016) already forecast the lower tail of IP shocks with 19 systemic-risk measures.

A structural model would tell the author which characteristics to use, and why.